

Universality and the right-angled Coxeter envelope of orthoscheme reflection groups

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Abstract

The right-triangle orbit-growth universality of the companion papers is the planar case of a general phenomenon. We prove an abstract dichotomy: for any family of quotients of a common group, spherical growth is shape-independent through a horizon equal to half the shortest added relator and first deviates exactly there. For orthoscheme (path-simplex) reflection groups the common group is the *right-angled Coxeter group* (RACG) on the orthoscheme's orthogonality graph, whose growth we compute in closed form: it is rational, with rate the algebraic number $r_n = 1 + 2 \cos \frac{2\pi}{n+3}$ (the spectral radius attached to the path P_{n+1}) and numerator $(1+t)^{\lfloor n/2 \rfloor + 1}$. The actual geometric group $W_T \leq \text{Isom}(\mathbb{R}^n)$ is *amenable*, hence a *proper* quotient of the non-amenable RACG, so its growth matches this rational envelope only up to a finite, shape-dependent depth and then deviates strictly below. In the plane the deviation occurs at depth 10, where the growth becomes the transcendental sequence A396406; in every higher dimension it lies beyond direct computation. We are therefore *not* entitled to conclude that higher dimensions are rational: the finite envelope agreement observed for $n \geq 3$ is the same pattern that, at $n = 2$, precedes the transcendental collapse. Whether the plane is genuinely exceptional, or every dimension is eventually transcendental, is left as an explicit open problem.

1 The abstract universality–deviation dichotomy

Let $G = \langle S \rangle$ be a group with finite symmetric generating set S and word length ℓ_G . For $N \trianglelefteq G$ write $Q = G/N$, $s_d(H) = \#\{h : \ell_H(h) = d\}$, and $K(N) = \min\{\ell_G(g) : g \in N \setminus \{1\}\}$.

Theorem 1 (Universality–deviation dichotomy). *With $n := \lceil K(N)/2 \rceil$: $s_d(Q) = s_d(G)$ for $d < n$, and $s_n(Q) < s_n(G)$.*

Proof. $\pi : G \rightarrow Q$ is surjective and 1-Lipschitz, so $s_d(Q) \leq s_d(G)$. If $d < n$ and $\pi g = \pi h$ ($g, h \in B_d(G)$) then $gh^{-1} \in N$ has length $\leq 2d < K(N)$, forcing $g = h$; the same bound excludes a shorter representative, so π is length-preserving on $B_d(G)$ and $s_d(Q) = s_d(G)$ for $d < n$. Splitting a geodesic word for $g_0 \in N$ with $\ell_G(g_0) = K(N)$ as $g_0 = w_1 w_2$, $\ell_G(w_1) = n$, gives two distinct elements of $B_n(G)$ sharing a fiber, so $s_n(Q) < s_n(G)$. $\square$

Applied to a family $\{Q_T = G/N_T\}$, this is universality: all members share G 's growth through their horizon $n_T = \lceil K(N_T)/2 \rceil$, then deviate. **The point of this paper is that for orthoscheme reflection groups the ambient G is a rational-growth RACG, but the members Q_T are proper quotients, so the shared rational growth is only a finite-depth envelope.**

2 The orthoscheme family and its RACG envelope

An n -orthoscheme has vertices $P_0, \dots, P_n$ with mutually orthogonal consecutive edges $P_{k-1}P_k = \ell_k e_k$; its $n+1$ facet reflections $R_0, \dots, R_n$ generate $W_T \leq \text{Isom}(\mathbb{R}^n)$ (the right-triangle group when

$n = 2$). Two facts hold for *every* orthoscheme: $R_i^2 = 1$, and $R_i R_j = R_j R_i$ when $|i - j| \geq 2$ (orthogonal facets). For generic legs consecutive facets meet at an irrational multiple of π , imposing no finite bond. The largest group compatible with these relations is the *right-angled Coxeter group*

$$W_n = \langle R_0, \dots, R_n \mid R_i^2 = 1, (R_i R_j)^2 = 1 \ (|i - j| \geq 2) \rangle,$$

and there is a surjection $\rho: W_n \twoheadrightarrow W_T$.

Theorem 2 (The RACG envelope is rational). *The spherical growth series of W_n is rational,*

$$W_n(t) = \frac{(1+t)^{\lfloor n/2 \rfloor + 1}}{N_n(t)}, \quad N_n(t) = (1+t)^{n+1} c_{\Gamma_n}\left(\frac{-1}{1+t}\right),$$

where c_{Γ_n} is the clique polynomial of the orthogonality graph Γ_n (vertices $0, \dots, n$; edges $|i - j| \geq 2$). Its growth rate is $r_n = 1 + 2 \cos \frac{2\pi}{n+3}$, the largest root of the reciprocal of N_n , and equals 1 plus the spectral radius $2 \cos \frac{2\pi}{n+3}$ of the path P_{n+1} . Explicitly

n	$W_n(t)$	r_n
2	$(1+t)^2/(1-t-t^2)$	$\varphi = 1.618\dots$
3	$(1+t)^2/(1-2t)$	2
4	$(1+t)^3/(1-2t-t^2+t^3)$	$1 + 2 \cos \frac{2\pi}{7} = 2.24698\dots$
5	$(1+t)^3/(1-3t+t^2+t^3)$	$1 + \sqrt{2}$
6	$(1+t)^4/(1-3t+3t^3)$	$1 + 2 \cos \frac{2\pi}{9} = 2.53209\dots$
7	$(1+t)^4/(1-2t-3t^2+4t^3-t^4)$	$\varphi^2 = 2.61803\dots$

Proof. W_n is a RACG, so its finite parabolics are exactly the cliques of Γ_n , each a group $(\mathbb{Z}/2)^{|C|}$ with polynomial growth $(1+t)^{|C|}$. Steinberg's formula gives $1/W_n(t^{-1}) = \sum_C \text{clique} (-1)^{|C|}/(1+t)^{|C|} = c_{\Gamma_n}(-1/(1+t))$; inverting and substituting $t \mapsto t^{-1}$ yields the displayed rational form. Γ_n is the complement of the path P_{n+1} , and the reciprocal of N_n is (a Chebyshev multiple of) the characteristic polynomial of P_{n+1} , whose largest eigenvalue is $2 \cos \frac{\pi}{n+2}$; a short computation with the independence polynomial of the path pins the dominant growth root at $r_n = 1 + 2 \cos \frac{2\pi}{n+3}$. The tabulated entries are the cases $n = 2, \dots, 7$, verified symbolically. $\square$

3 The geometric group is a proper amenable quotient

Proposition 3 (Finite-horizon envelope). *For every $n \geq 2$, W_T is a proper quotient of W_n ; consequently, by Theorem 1, the geometric growth $s_d(W_T)$ equals the rational envelope $[t^d]W_n(t)$ only for $d < n_T := \lceil K(\ker \rho)/2 \rceil$, and $s_{n_T}(W_T) < [t^{n_T}]W_n(t)$.*

Proof. $W_T \leq \text{Isom}(\mathbb{R}^n) = \mathbb{R}^n \rtimes O(n)$ is amenable (an extension of the amenable $\mathbb{R}^n$ by the compact $O(n)$). The RACG W_n is non-amenable: its orthogonality graph Γ_n is not complete multipartite (its complement, the path P_{n+1} , is connected and not a disjoint union of cliques for $n \geq 2$), so W_n contains a free subgroup of rank 2. An amenable group cannot be isomorphic to a non-amenable one, so $\ker \rho \neq \{1\}$ and W_T is a proper quotient. Theorem 1 then gives the horizon. $\square$

Corollary 4 (The plane). *For $n = 2$ the horizon is $n_T = 10$: $s_d(W_T) = F_{d+3} = [t^d]W_2(t)$ for $d \leq 9$, and at $d = 10$ the growth deviates, becoming the sequence A396406, which is transcendental over $\mathbb{Q}$ (companion paper: the relaxed companion unconditionally; the true series conditionally on one amplitude estimate) with rate $\beta_2 = 1.4916\dots < \varphi = r_2$. This is the only dimension in which the deviation has been computed past its onset.*

4 What is proved, and the open problem

Two things are proved unconditionally and dimension-freely: the universality–deviation dichotomy (Theorem 1) and the rationality of the RACG envelope with rate r_n (Theorem 2). A third is proved structurally: the geometric group is always a proper amenable quotient, so its growth is the envelope only up to a finite horizon (Proposition 3). For $3 \leq n \leq 7$ the geometric growth was computed to depth 13–16 and agrees with the envelope throughout that range; but Proposition 3 guarantees it must eventually deviate below, and the deviation depth n_T for $n \geq 3$ exceeds what we can reach.

The tempting conclusion — that the geometric growth is rational for $n \geq 3$, so the plane is the unique transcendental dimension — is *not* justified. The evidence for it is precisely the finite envelope agreement, and at $n = 2$ that same agreement (through depth 9) precedes the transcendental collapse. We therefore state the honest alternative as an open problem rather than a theorem.

Question 5 (Eventual transcendence). For $n \geq 3$, is the geometric growth series of the generic n -orthoscheme group rational, algebraic, or transcendental? Equivalently: what is the arithmetic of the amenable quotient $W_n/\ker \rho$ past its deviation horizon n_T ? The plane is exceptional iff the answer is “rational” for all $n \geq 3$; the alternative — eventual transcendence in every dimension, with the plane merely the earliest and only computable case — is equally consistent with all current evidence.

The companion papers are recast accordingly: their object, the planar right-triangle group, is the one dimension in which the amenable quotient of the Coxeter envelope has been carried past its deviation and found transcendental. Whether that transcendence is a two-dimensional accident or the first instance of a universal phenomenon is open.